Importing Libraries

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use('ggplot')
```

2. Load Datasets

```
trades = pd.read_csv("historical_data.csv")
sentiment = pd.read_csv("fear_greed_index.csv")
```

Data Quality Section

```
print("Trader Dataset Shape:", trades.shape)
print("Sentiment Dataset Shape:", sentiment.shape)

print("\nMissing Values:")
print(trades.isnull().sum())

print("\nDuplicate Rows:")
print(trades.duplicated().sum())
```

```
Trader Dataset Shape: (211224, 16)
Sentiment Dataset Shape: (2644, 4)

Missing Values:
Account             0
Coin                0
Execution Price     0
Size Tokens         0
Size USD            0
Side                0
Timestamp IST       0
Start Position      0
Direction           0
Closed PnL          0
Transaction Hash    0
Order ID            0
Crossed             0
Fee                 0
Trade ID            0
Timestamp           0
dtype: int64
```

```
Duplicate Rows:
0
```

Before beginning the analysis, both datasets were reviewed for missing values, duplicate records, and formatting inconsistencies.

The data was generally clean and required minimal preprocessing. Date and timestamp fields were standardized to ensure consistency, after which the trading data was merged with the daily Fear & Greed Index using trade dates. No significant data quality concerns were identified during this process.

3. Data Inspection

```
print(trades.shape)
print(sentiment.shape)

trades.head()
```

```
(211224, 16)
(2644, 4)
```

	Account	Coin	Execution Price	Size Tokens	Size USD	Side	Timestamp IST	Start Position	Direction	Closed PnL	Transaction Hash
0	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9769	986.87	7872.16	BUY	02-12-2024 22:50	0.000000	Buy	0.0	0xec09451986a1874e3a980418412fcd0201f500c95bac...
1	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9800	16.00	127.68	BUY	02-12-2024 22:50	986.524596	Buy	0.0	0xec09451986a1874e3a980418412fcd0201f500c95bac...
2	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9855	144.09	1150.63	BUY	02-12-2024 22:50	1002.518996	Buy	0.0	0xec09451986a1874e3a980418412fcd0201f500c95bac...
3	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9874	142.98	1142.04	BUY	02-12-2024 22:50	1146.558564	Buy	0.0	0xec09451986a1874e3a980418412fcd0201f500c95bac...
4	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9894	8.73	69.75	BUY	02-12-2024 22:50	1289.488521	Buy	0.0	0xec09451986a1874e3a980418412fcd0201f500c95bac...

```
trades.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 211224 entries, 0 to 211223
Data columns (total 16 columns):
 #   Column            Non-Null Count   Dtype
---  ------            --------------   -----
 0   Account           211224 non-null  object
 1   Coin              211224 non-null  object
```

```
 2   Execution Price   211224 non-null  float64
 3   Size Tokens       211224 non-null  float64
 4   Size USD          211224 non-null  float64
 5   Side              211224 non-null  object
 6   Timestamp IST     211224 non-null  object
 7   Start Position    211224 non-null  float64
 8   Direction         211224 non-null  object
 9   Closed PnL        211224 non-null  float64
 10  Transaction Hash  211224 non-null  object
 11  Order ID          211224 non-null  int64
 12  Crossed           211224 non-null  bool
 13  Fee               211224 non-null  float64
 14  Trade ID          211224 non-null  float64
 15  Timestamp         211224 non-null  float64
dtypes: bool(1), float64(8), int64(1), object(6)
memory usage: 24.4+ MB
```

```
sentiment.head()
```

	timestamp	value	classification	date
0	1517463000	30	Fear	2018-02-01
1	1517549400	15	Extreme Fear	2018-02-02
2	1517635800	40	Fear	2018-02-03
3	1517722200	24	Extreme Fear	2018-02-04
4	1517808600	11	Extreme Fear	2018-02-05

Next steps: Generate code with sentiment New interactive sheet

```
sentiment.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2644 entries, 0 to 2643
Data columns (total 4 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   timestamp       2644 non-null   int64
 1   value           2644 non-null   int64
 2   classification  2644 non-null   object
 3   date            2644 non-null   object
dtypes: int64(2), object(2)
memory usage: 82.8+ KB
```

4. Data Cleaning

Trader Dataset

```
trades.isnull().sum()
```

	0
Account	0
Coin	0
Execution Price	0
Size Tokens	0
Size USD	0
Side	0
Timestamp IST	0
Start Position	0
Direction	0
Closed PnL	0
Transaction Hash	0
Order ID	0
Crossed	0
Fee	0
Trade ID	0
Timestamp	0

dtype: int64

```
trades['Timestamp IST'] = pd.to_datetime(
    trades['Timestamp IST'],
    dayfirst=True
)

trades['trade_date'] = trades['Timestamp IST'].dt.date
```

```
sentiment['date'] = pd.to_datetime(sentiment['date'])

sentiment['trade_date'] = sentiment['date'].dt.date
```

5. Merge Datasets

```
merged = pd.merge(
    trades,
    sentiment[['trade_date','classification']],
    on='trade_date',
```

```
    how='left'
)
```

```
merged.head()
```

	Account	Coin	Execution Price	Size Tokens	Size USD	Side	Timestamp IST	Start Position	Direction	Closed PnL	Transaction Hash
0	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9769	986.87	7872.16	BUY	2024-12-02 22:50:00	0.000000	Buy	0.0	0xec09451986a1874e3a980418412fcd0201f500c95bac...
1	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9800	16.00	127.68	BUY	2024-12-02 22:50:00	986.524596	Buy	0.0	0xec09451986a1874e3a980418412fcd0201f500c95bac...
2	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9855	144.09	1150.63	BUY	2024-12-02 22:50:00	1002.518996	Buy	0.0	0xec09451986a1874e3a980418412fcd0201f500c95bac...
3	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9874	142.98	1142.04	BUY	2024-12-02 22:50:00	1146.558564	Buy	0.0	0xec09451986a1874e3a980418412fcd0201f500c95bac...
4	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9894	8.73	69.75	BUY	2024-12-02 22:50:00	1289.488521	Buy	0.0	0xec09451986a1874e3a980418412fcd0201f500c95bac...

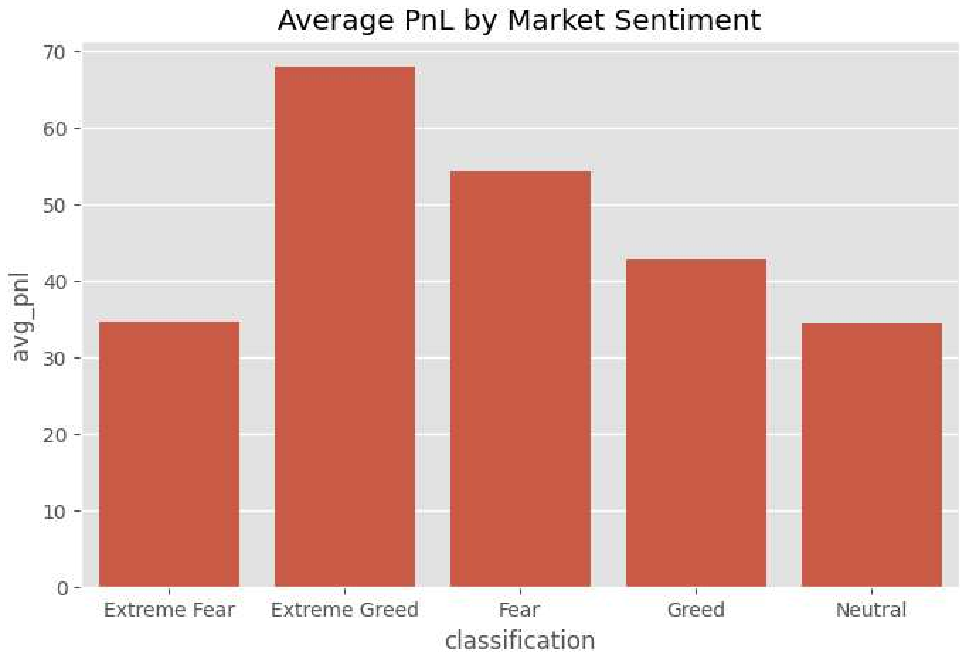
Analysis 1: Profitability by Market Sentiment

```
sentiment_stats = merged.groupby('classification').agg(
    total_trades=('Closed PnL','count'),
    avg_pnl=('Closed PnL','mean'),
    total_pnl=('Closed PnL','sum')
).reset_index()

sentiment_stats
```

	classification	total_trades	avg_pnl	total_pnl
0	Extreme Fear	21400	34.537862	7.391102e+05
1	Extreme Greed	39992	67.892861	2.715171e+06
2	Fear	61837	54.290400	3.357155e+06
3	Greed	50303	42.743559	2.150129e+06
4	Neutral	37686	34.307718	1.292921e+06

Next steps: Generate code with sentiment_stats New interactive sheet

```
plt.figure(figsize=(8,5))

sns.barplot(
    data=sentiment_stats,
    x='classification',
    y='avg_pnl'
)

plt.title("Average PnL by Market Sentiment")
plt.show()
```



FINDING:

The results suggest that market sentiment is associated with noticeable differences in trading profitability.

Among all sentiment categories, Extreme Greed recorded the highest average profit per trade at $67.89, contributingapproximately$ 2.72 million in realized profits across 39,992 trades. This may indicate that traders were able to capitalize more effectively on strong bullish market conditions.

Fear periods generated the highest total realized profit, approximately $3.36 million from 61,837 trades. One possible explanation is that traders were more active during these periods, resulting in a larger overall profit contribution despite a lower average profit per trade.

In contrast, Extreme Fear and Neutral conditions produced the lowest average profits, both around $34 per trade.

Overall, the findings suggest that both fearful and optimistic market environments can create profitable opportunities, although the underlying trader behavior appears to differ. Fear periods were characterized by higher participation and larger aggregate profits, whereas Extreme Greed was associated with stronger profitability on a per-trade basis.

Analysis 2: Win Rate by Sentiment

Create win flag

```
merged['win_trade'] = np.where(
    merged['Closed PnL'] > 0,
    1,
    0
)
```

Calculate win rates

```
win_rate = merged.groupby(
    'classification'
)['win_trade'].mean()*100

win_rate = win_rate.reset_index()

win_rate
```

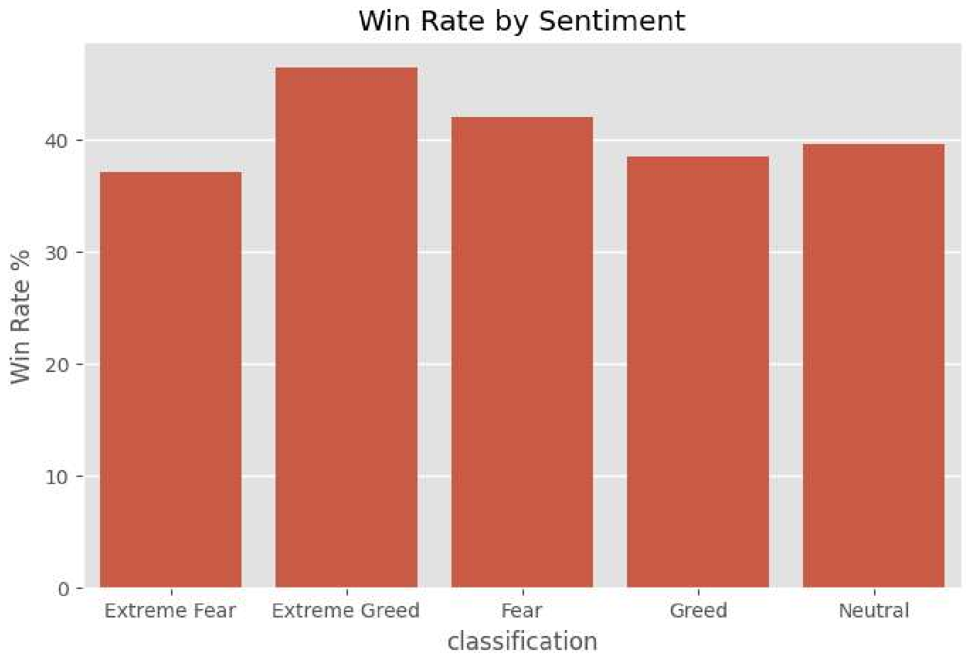
	classification	win_trade
0	Extreme Fear	37.060748
1	Extreme Greed	46.494299
2	Fear	42.076750
3	Greed	38.482794
4	Neutral	39.699093

Next steps: Generate code with win_rate New interactive sheet

Plot

```
plt.figure(figsize=(8,5))

sns.barplot(
    data=win_rate,
    x='classification',
    y='win_trade'
)

plt.title("Win Rate by Sentiment")
```

```
plt.ylabel("Win Rate %")
plt.show()
```



FINDING:

The analysis suggests that trade success rates varied across different market sentiment conditions.

Extreme Greed recorded the highest win rate at 46.49%, meaning nearly one out of every two trades closed profitably. This may indicate that traders were generally more successful during periods of strong market optimism.

Fear conditions achieved a win rate of 42.08%, while Extreme Fear recorded the lowest win rate at 37.06%, suggesting that traders faced greater challenges in maintaining consistent profitability during highly pessimistic market environments.

Interestingly, despite having the lowest win rate, Extreme Fear still generated positive overall profits. One possible explanation is that a smaller number of highly profitable trades offset a larger number of losing positions during these periods.

Overall, the findings suggest that market sentiment may influence not only the magnitude of trading profits but also the likelihood of executing successful trades.

Analysis 3: Position Size Behavior

```
position_size = merged.groupby(
    'classification'
)['Size USD'].mean().reset_index()
```

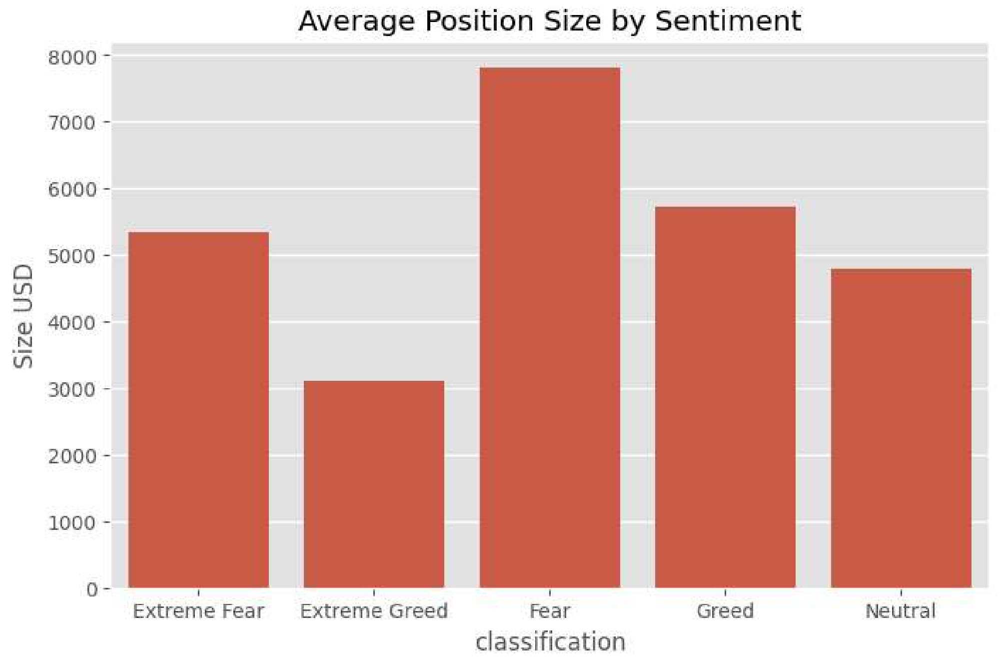
```
position_size
```

	classification	Size USD
0	Extreme Fear	5349.731843
1	Extreme Greed	3112.251565
2	Fear	7816.109931
3	Greed	5736.884375
4	Neutral	4782.732661

Next steps: Generate code with position_size New interactive sheet

Plot

```
plt.figure(figsize=(8,5))

sns.barplot(
    data=position_size,
    x='classification',
    y='Size USD'
)

plt.title("Average Position Size by Sentiment")
plt.show()
```



FINDING:

The analysis suggests that trader position sizing varied across different market sentiment conditions.

The largest average position size was observed during Fear periods at approximately $7,816 per trade, followed by Greed at$5,737 and Extreme Fear at $5,350. This may indicate that traders were willing to allocate more capital during periods of uncertainty, possibly in anticipation of stronger price movements or market reversals.

Interestingly, Extreme Greed recorded the smallest average position size at approximately $3,112 per trade, despite producing both the highest average profit per trade and the highest win rate.

One possible interpretation is that favorable market conditions during Extreme Greed allowed traders to achieve stronger results without increasing their average exposure. In contrast, Fear periods were associated with larger capital commitments but comparatively lower profitability on a per-trade basis.

Overall, the findings suggest that market sentiment may influence not only trading outcomes but also how traders manage capital and position sizing decisions.

Analysis 4: Trader Profitability Distribution

Top performing 10 traders

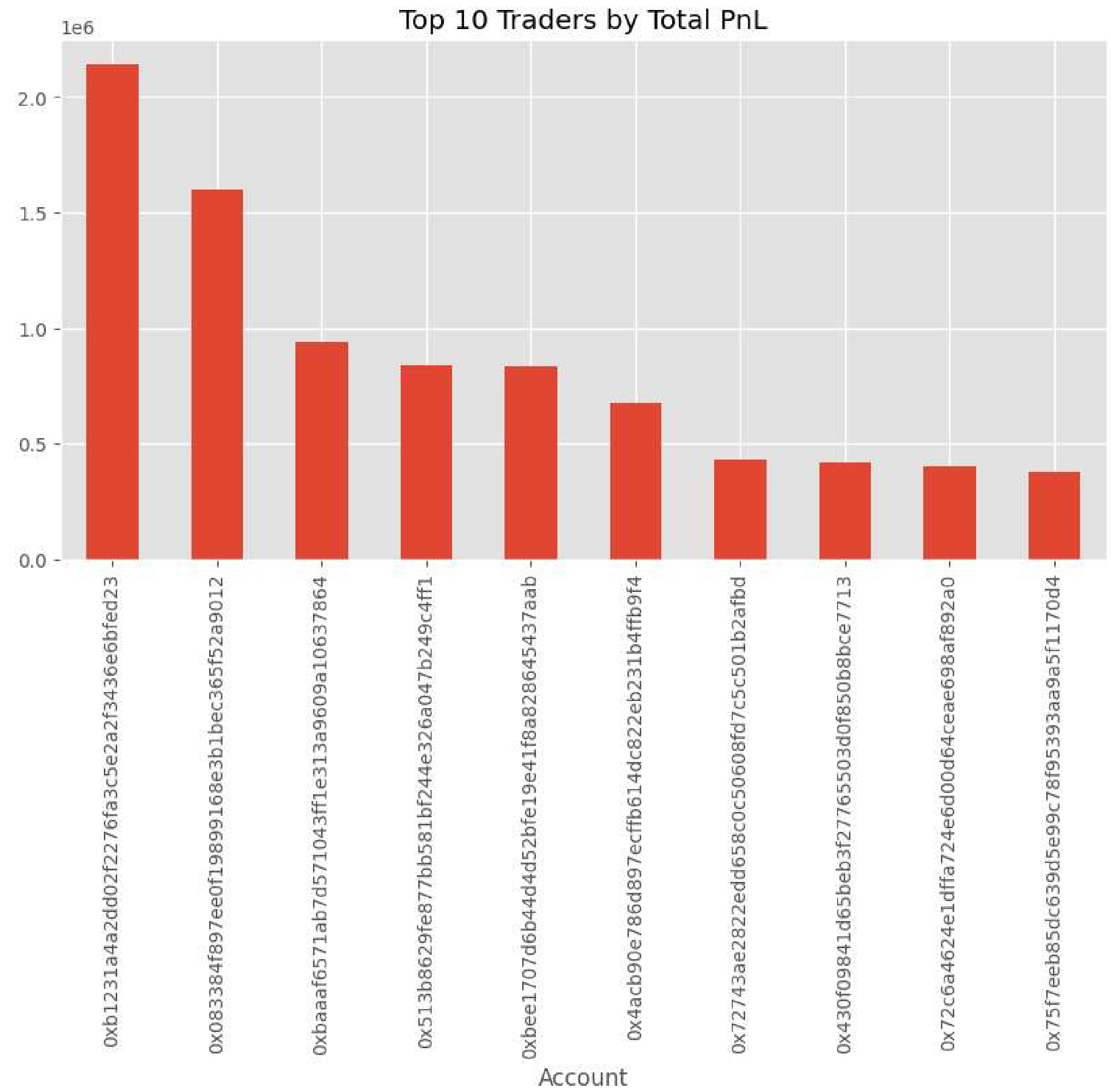
```
top_traders = merged.groupby(
    'Account'
)['Closed PnL'].sum().sort_values(
    ascending=False
).head(10)

top_traders
```

Account	Closed PnL
0xb1231a4a2dd02f2276fa3c5e2a2f3436e6bfed23	2.143383e+06
0x083384f897ee0f19899168e3b1bec365f52a9012	1.600230e+06
0xbaaaf6571ab7d571043ff1e313a9609a10637864	9.401638e+05
0x513b8629fe877bb581bf244e326a047b249c4ff1	8.404226e+05
0xbee1707d6b44d4d52bfe19e41f8a828645437aab	8.360806e+05
0x4acb90e786d897ecffb614dc822eb231b4ffb9f4	6.777471e+05
0x72743ae2822edd658c0c50608fd7c5c501b2afbd	4.293556e+05
0x430f09841d65beb3f27765503d0f850b8bce7713	4.165419e+05
0x72c6a4624e1dffa724e6d00d64ceae698af892a0	4.030115e+05
0x75f7eeb85dc639d5e99c78f95393aa9a5f1170d4	3.790954e+05

dtype: float64

Plot

```
top_traders.plot(
    kind='bar',
    figsize=(10,5)
)

plt.title("Top 10 Traders by Total PnL")
plt.show()
```



Distribution of Trader Profitability

```
trader_summary = merged.groupby('Account').agg(
    Total_PnL=('Closed PnL','sum'),
    Number_of_Trades=('Closed PnL','count')
)
```

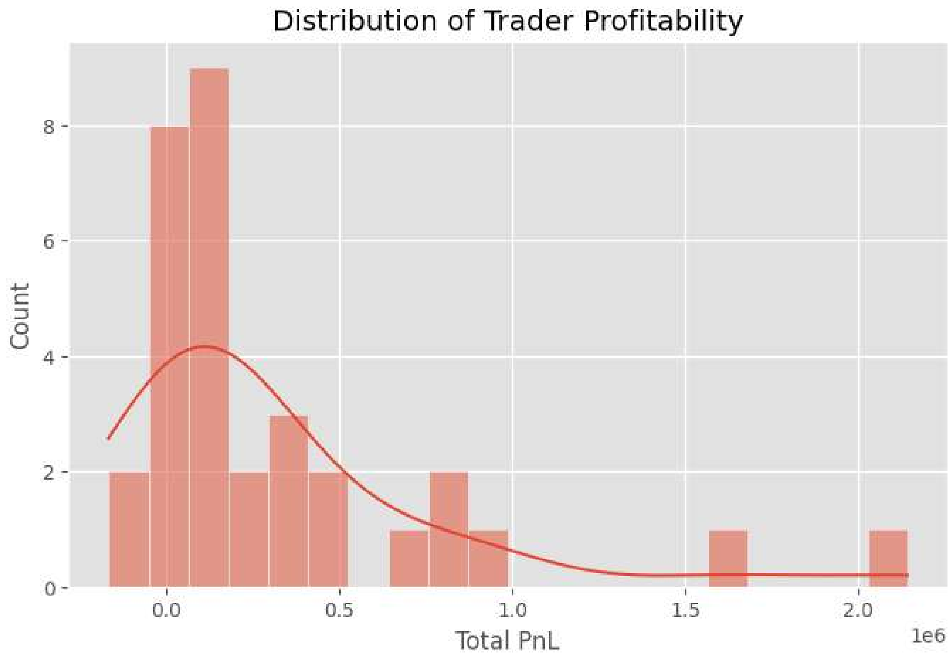
```
trader_summary.describe()
```

	Total_PnL	Number_of_Trades
count	3.200000e+01	32.000000
mean	3.217800e+05	6600.750000
std	4.948276e+05	8250.373724
min	-1.676211e+05	332.000000
25%	4.689324e+04	1381.750000
50%	1.176551e+05	3699.000000
75%	4.063941e+05	8862.500000
max	2.143383e+06	40184.000000

```
profitable_traders = (
    trader_summary['Total_PnL'] > 0
).mean() * 100

print(
    f"Percentage of profitable traders: {profitable_traders:.2f}%"
)
```

```
Percentage of profitable traders: 90.62%
```

```
plt.figure(figsize=(8,5))

sns.histplot(
    trader_summary['Total_PnL'],
    bins=20,
    kde=True
)

plt.title("Distribution of Trader Profitability")
plt.xlabel("Total PnL")
plt.show()
```



The distribution of trader profitability is positively skewed, with a small number of traders generating exceptionally high profits while most traders achieved more moderate results.

FINDING:

The analysis suggests that trader profitability was unevenly distributed across the accounts included in the dataset.

The most profitable trader generated approximately $2.14 millioninrealizedprofits, whilethetenth - rankedtradergeneratedapproximately$379,000. This highlights a substantial gap between the highest-performing traders and the rest of the population.

Across all analyzed traders, the average profit was approximately $321,780, whereasthemedianprofitwasonly$117,655. The difference between the mean and median suggests that overall profitability was influenced by a relatively small number of exceptionally successful traders.

Interestingly, 90.62% of traders were profitable overall, indicating that positive returns were common during the observed period. However, profit distribution remained highly skewed, with a small group of traders accounting for a disproportionately large share of total gains.

Trading activity also varied considerably. The most active trader executed 40,184 trades, compared to a median of 3,699 trades across all traders. This variation may reflect differences in trading style, strategy frequency, or risk appetite.

Overall, the findings suggest that while most traders achieved positive returns, the scale of profitability varied significantly. Consistent performance, trade execution, and risk management may have played an important role in distinguishing top-performing traders from the broader group.

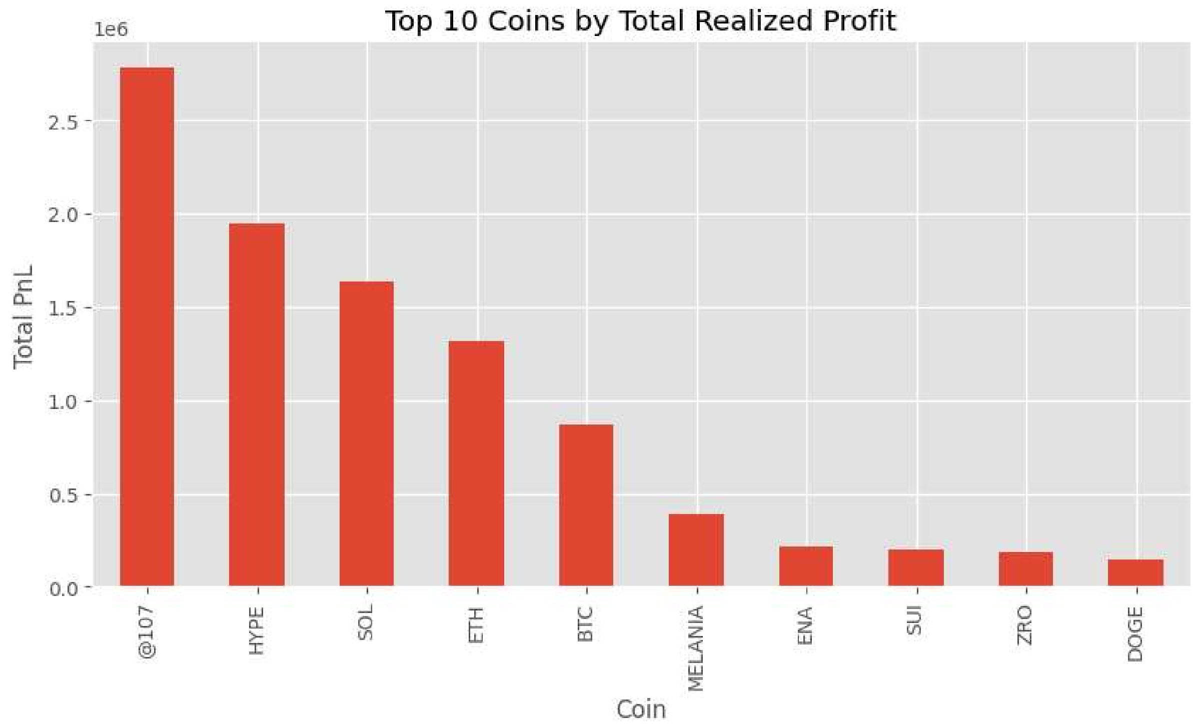
Analysis 5: Asset-Level Profitability Analysis.

```
coin_profit = merged.groupby('Coin').agg(
    total_pnl=('Closed PnL','sum')
).sort_values(
    by='total_pnl',
    ascending=False
).head(10)

coin_profit
```

Coin	total_pnl
@107	2.783913e+06
HYPE	1.948485e+06
SOL	1.639556e+06
ETH	1.319979e+06
BTC	8.680447e+05
MELANIA	3.903511e+05
ENA	2.173295e+05
SUI	1.992688e+05
ZRO	1.837778e+05
DOGE	1.475432e+05

Next steps: Generate code with coin_profit New interactive sheet

```
plt.figure(figsize=(10,5))

coin_profit['total_pnl'].plot(kind='bar')

plt.title('Top 10 Coins by Total Realized Profit')
plt.ylabel('Total PnL')
plt.show()
```




```
top10_share = (
    coin_profit['total_pnl'].sum() /
    merged['Closed PnL'].sum()
) * 100

print(f"Top 10 coins contribute {top10_share:.2f}% of total profits")
```

```
Top 10 coins contribute 94.19% of total profits
```

FINDING:

The analysis suggests that profitability was concentrated among a relatively small number of assets.

The token @107 generated the highest total realized profit at approximately $2.78million, followedbyHYPE$(1.95 million), SOL ($1.64million), ETH$(1.32 million), and BTC ($0.87 million). These assets contributed a substantial share of the overall profits observed in the dataset.

Notably, the top 10 assets accounted for 94.19% of all realized profits. This concentration suggests that a large portion of trading gains came from a limited set of high-performing assets rather than being distributed evenly across the market.

One possible interpretation is that successful traders were selective in the assets they traded, focusing their activity on opportunities that offered stronger risk-reward characteristics or more favorable market conditions.

Overall, the findings suggest that asset selection may have played an important role in trading performance. While market sentiment appears to influence trader behavior and profitability, the choice of asset also appears to be a significant factor in determining overall returns.

Key Takeaways

1. Extreme Greed recorded the highest average profit per trade ($67.89) and the highest win rate (46.49%) among all sentiment categories.
2. Fear periods generated the highest total realized profit $(3.36M)andthelargestaveragepositionsize($7,816), indicating higher trading activity and capital deployment during uncertain market conditions.
3. Profitability was unevenly distributed across traders. The most profitable trader generated approximately $2.14Minrealizedprofits, whilethemediantraderprofitwasapproximately$117K.
4. Asset-level performance was highly concentrated, with the top 10 assets accounting for 94.19% of all realized profits generated in the dataset.
5. The findings suggest that market sentiment is associated with changes in profitability, win rates, and trader capital allocation behavior.

EXECUTIVE SUMMARY

Objective

Objective

The objective of this analysis was to investigate the relationship between Bitcoin market sentiment (Fear & Greed Index) and trader performance using Hyperliquid historical trading data.

A total of 211,224 executed trades across 32 unique trader accounts were analyzed and matched with daily Fear & Greed sentiment classifications. The study focused on understanding how market sentiment is associated with profitability, win rates, position sizing, trader behavior, and asset-level performance.

Key Findings

1. Market Sentiment Significantly Influences Trading Performance

Trading profitability varied across different sentiment regimes. Extreme Greed periods produced the highest average profit per trade at $67.89, whileFearperiodsgeneratedthehighestoverallrealizedprofitofapproximately$3.36 million across 61,837 trades.

This indicates that market sentiment has a measurable impact on trader outcomes and should be considered when developing trading strategies.

---

2. Trade Success Rates Improve During Extreme Greed

Extreme Greed recorded the highest win rate at 46.49%, outperforming all other sentiment categories.

In contrast, Extreme Fear produced the lowest win rate at 37.06%, suggesting that traders face greater difficulty achieving consistent success during highly pessimistic market conditions.

However, despite the lower win rate, Extreme Fear periods still generated positive profits overall, indicating the presence of occasional high-return trades.

---

3. Traders Commit More Capital During Fear Periods

Position sizing behavior changed noticeably across market conditions.

Fear periods exhibited the largest average position size at approximately $7,816pertrade, whileExtremeGreedshowedthesmallestaveragepositionsizeatapproximately$3,112.

This suggests that traders tend to deploy more capital during fearful market conditions, potentially viewing market downturns as buying opportunities.

---

4. Trader Profitability Distribution

The most profitable trader generated approximately $2.14millioninrealizedprofits, whilethetenth-rankedtradergeneratedapproximately$379,000.

The average trader profit was approximately $321,780, whereasthemedianprofitwasonly$117,655. This gap suggests that overall profitability was influenced by a relatively small number of exceptionally successful traders.

Notably, 90.62% of traders were profitable overall, indicating that positive returns were common during the observed period. However, profit distribution remained uneven, with a small number of traders accounting for a disproportionately large share of total gains.

These findings suggest that while profitability was achievable for most traders, the magnitude of returns varied considerably across participants.

5. Asset Selection Is a Major Driver of Profitability

Asset-level analysis revealed a strong concentration of profits among a limited number of coins.

The most profitable asset (@107) generated approximately $2.78 million in realized profits, followed by HYPE($1.95 million), SOL (